

**JOYCE J.Y. LIN, PhD**  
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## EDUCATION AND TRAINING

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- 2024 – Postdoctoral Fellow, Environmental Health**  
Harvard T.H. Chan School of Public Health, Boston, MA  
NIH/NIEHS K99/R00 Pathway to Independence Awardee  
Mentor: Marc Weisskopf
- 2019 – 2024 Doctor of Philosophy (PhD), Environmental Health and Engineering**  
Johns Hopkins Bloomberg School of Public Health, Baltimore, MD  
Dissertation title: Metal biomonitoring in low mass toenail samples: applications in environmental epidemiology
- 2018 – 2019 Master of Health Science (MHS), Environmental Health and Engineering**  
Johns Hopkins Bloomberg School of Public Health, Baltimore, MD  
*Specialization in Environmental Epidemiology*  
*Certificate in Risk Sciences and Public Policy*
- 2014 – 2018 Bachelor of Arts (BA), Public Health Studies**  
Johns Hopkins University, Krieger School of Arts and Sciences, Baltimore, MD

## RESEARCH SUPPORT

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### Current:

**NIH/NIEHS K99/R00 Pathway to Independence Award (K99ES038596)**  
Project Period: July 1, 2026 – June 30, 2028  
Project Title: Early Life Lead Exposure and Later Life Cognitive Function: Biological and Psychosocial Pathways  
Role: Principal Investigator

### Completed:

**NIOSH ERC Pilot Project Research Training Award**  
Project Period: 06/2021 – 06/2023  
Project Title: Characterizing Chronic Metal Exposure in the US Gulf States: Qualification of a Toenail Biomarker and Implications for Neurobehavioral Health  
Role: Principal Investigator

**JHU 21<sup>st</sup> Century Cities Award for Doctoral Research on Urban Issues**  
Project Period: 06/2021 - 06/2022  
Project Title: Characterizing Chronic Metal Exposure and Implications for Neurobehavioral Health in the US Gulf States  
Role: Principal Investigator

**JHU Urban Health Institute Small Grant Award**

Project Period: 01/2017 – 01/2018

Project Title: Central Baltimore Community Health Strategy

Role: Principal Investigator

**PEER REVIEW PUBLICATIONS**

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Published

*\*Indicates corresponding author*

1. Leung M, Farmer JG, Tang IW, **Lin JJY**, Mucci L, Mittleman M, *et al.* Mortality and Childhood Residence Near the Radiologically Contaminated Coldwater Creek, St. Louis, Missouri. *Int J Epidemiol.* Accepted.
2. **Lin JJY\***, Hickman R, Leung M, Farmer J, Tang IW, McAlaine K, *et al.* Prenatal and early postnatal lead exposure and later adulthood BMI: The Saint Louis Baby Tooth–Later Life Health Study. *Environ Res.* 2026;302:124606.
3. Leung M, **Lin JJY**, Mittleman MA, Farmer JG, Tang IW, McAlaine K, *et al.* Childhood residential proximity to the radiologically contaminated Coldwater Creek and later-life cardiovascular disease in the St. Louis Baby Tooth Study. *Sci Total Environ.* 2026;1044:181957.
4. Hu B, **Lin JJY**, Qiu X, McAlaine K, Mahalingaiah S, Weisskopf MG. Sleep disturbances during perimenopause and later-life cognitive function. *Menopause.* Published online June 16, 2026. doi:10.1097/GME.0000000000002790.
5. Hickman RC, **Lin JJY**, McAlaine KA, Farmer JG, Tang IW, Leung M, *et al.* Prenatal and Early Postnatal Lead Exposure and Later Adulthood Cognitive Function in the St. Louis Baby Tooth–Later Life Health Study. *Neurology.* 2026;106(6):e214616.
6. **Lin JJY**, Hickman R, Farmer J, Leung M, Tang IW, McAlaine K, *et al.* Prenatal and Early Postnatal Lead Exposure, Sensitive Periods, and Later Adult Mental Health. *JAMA Psychiatry.* 2025;82(12):1225–1233.
7. Leung M, Tang IW, **Lin JJY**, Mucci L, Farmer JG, McAlaine K, *et al.* Cancer Incidence and Childhood Residence Near the Coldwater Creek Radioactive Waste Site. *JAMA Netw Open.* 2025;8(7):e2521926.
8. **Lin JJY\***, Kuiper JR, Dickerson AS, Buckley JP, Volk HE, Rohlman DS, *et al.* Associations of a toenail metal mixture with attention and memory in the Gulf Long-term Follow-up (GuLF) Study. *Sci Total Environ.* 2024;935:173387.
9. **Lin JJY\***, Werder EJ, Lawrence KG, Braxton Jackson W, Sandler DP, Dickerson AS, *et al.* Residential Proximity to Metal-Emitting Industries and Toenail Metal Concentration in the US Gulf States. *Expo Health.* 2024.
10. **Lin JJY\***, Koffman LJ, Tehrani MW, Chen R, Han SG, Sandler DP, *et al.* Reliability of low-mass toenail samples as biomarkers of chronic metal exposure. *J Expo Sci Environ Epidemiol.* 2023;33:945–953.
11. Tillery A, Aherrera A, Chen R, **Lin JJY**, Tehrani M, Moustafa D, *et al.* Characterization of e-cigarette users according to device type, use behaviors, and self-reported health outcomes: Findings from the EMIT study. *Tob Induc Dis.* 2023;21:1–11.
12. Aherrera A, **Lin JJ**, Chen R, Tehrani M, Schultze A, Borole A, *et al.* Metal concentrations in e-cigarette aerosol samples: A comparison by device type and flavor. *Environ Health Perspect.* 2023;131(12):127004.

13. **Lin JJY**, Buehler C, Datta A, Gentner DR, Koehler K, Zamora ML. Laboratory and field evaluation of a low-cost methane sensor and key environmental factors for sensor calibration. *Environ Sci Atmos*. 2023;3(4):683–694.
14. **Lin JJY\***, Tehrani MW, Chen R, Heaney CD, Rule AM. Characterizing spatiotemporal variability in airborne heavy metal concentration: Changes after 18 years in Baltimore, MD. *Environ Res*. 2022;209:112878.

## SELECTED MEDIA AND RESEARCH HIGHLIGHTS

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Prenatal lead exposure associated with later-life cognitive function

- *Harvard T.H. Chan School of Public Health News*
- *American Academy of Neurology Press Release*
- *Medscape*

Prenatal lead exposure linked with increased risk of adult depression

- *Harvard T.H. Chan School of Public Health News*
- *Medscape*

Childhood residence near Coldwater Creek associated with later-life cancer risk

- *Harvard T.H. Chan School of Public Health News*
- *U.S. News & World Report*
- *Yahoo News*

## TEACHING

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Instructor of record

**Primary Instructor**, *Johns Hopkins University* “Urban Health: Global Perspectives for Sustainable Development” (AS.280.460) Fall 2021, Fall 2022

Guest Lectures

“**Chemical Hazards**” in “Environmental Studies Capstone”, *Brandeis University, Waltham, MA*. 2024

“**Gravimetric PM Monitoring**” in “Industrial Hygiene Laboratory” (182.614.01) *Johns Hopkins Bloomberg School of Public Health, Baltimore, MD*. 2021, 2022

“**Exposure Assessment**” in “Environmental Health” (180.601.61), *Johns Hopkins Bloomberg School of Public Health, Baltimore, MD*. 2020, 2021, 2022

Teaching Assistantships

**Section Leader**, *Johns Hopkins University* “Environment and Your Health” Introduction to Public Health

(AS.280.101) Fall 2020, 2021, Fall 2022

**Teaching Assistant**, *Johns Hopkins Bloomberg School of Public Health* “The Built Environment: Influences and Challenges to Improving Population Health” (180.625.01) Summer 2020, 2021, 2022

**Teaching Assistant**, *Johns Hopkins University* “Environment and Your Health” (AS.280.335) Fall 2020, 2021, Fall 2022

**Teaching Assistant, Johns Hopkins Bloomberg School of Public Health “Environmental Health”**  
(180.601.01) Spring 2021

**Teaching Assistant, Johns Hopkins Bloomberg School of Public Health “Community Driven Epidemiology and Environmental Justice”** (180.625.01) Fall 2020

**Teaching Assistant, Johns Hopkins Bloomberg School of Public Health “Methods in Quantitative Health Risk Assessment”** (317.605.81 ) Spring 2019

## RESEARCH MENTORSHIP

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### PhD students

Monique Slowly, Population Health Sciences program – Harvard T.H. Chan School of Public Health (2025 – present)

Marwa Osman, Biological Science in Public Health program – Harvard T.H. Chan School of Public Health (2024 – present)

### Master’s students

Esther Sun, MS in Environmental Health – Harvard T.H. Chan School of Public Health (2025 – present)

Hanyu Liu, MPH in Epidemiology – Harvard T.H. Chan School of Public Health (2025 – present)

### Undergraduate students

Arianna Kales – Boston College, RA at Harvard T.H. Chan School of Public Health (2024 – present)

Aghader Yassen – Harvard College, RA at Harvard T.H. Chan School of Public Health (2026– present)

Isabella Chavarro – William and Mary, RA at Harvard T.H. Chan School of Public Health (2024 – present)

Kevin Han – RA at Johns Hopkins University (2021 – 2023)

## HONORS AND AWARDS

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2026 NIH/NIEHS Pathway to Independence Award (K99/R00)

2025 Environmental Health Values Award, Harvard T.H. Chan School of Public Health

2023 Gordis Teaching Fellowship Award, Johns Hopkins University

2022 Millard Bass Research Fund Scholar, Johns Hopkins Bloomberg School of Public Health

2022 ISES Conference Award, International Society of Exposure Science

2022 Jane Warren Trainee Conference Award, Health Effects Institute

2022 Gordis Teaching Fellowship Award, Johns Hopkins University

2021 ISES Conference Award, International Society of Exposure Science

2019 Delta Omega, Alpha Chapter, Johns Hopkins Bloomberg School of Public Health

2018 Bloomberg Centennial Scholar, Johns Hopkins Bloomberg School of Public Health

2018 University Honors, Johns Hopkins University

## PRESENTATIONS

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### Invited Research Seminars

1. **Lin JJY**. Prenatal and Early Postnatal Lead Exposure, Sensitive Periods, and Later Adult Mental Health. Invited oral presentation, Harvard NIEHS Center for Environmental Health Research Retreat, Harvard T.H. Chan School of Public Health, Boston, MA. 2025.

### Scientific Meetings (\*Indicates primary presenter)

#### Oral Presentations

1. **Lin JJY\***, Weisskopf MG, The Saint Louis Baby Tooth - Later Life Health Study (SLBT): Early Life Exposures, Critical Windows, and Long-Term Health. Invited symposium presentation, *Decoding dementia: Environmental risk factors and mechanisms throughout the lifespan*, International Society of Environmental Epidemiology, North America Chapter Meeting, Baltimore Maryland. 2026
2. **Lin JJY\***, Hickman R, Farmer J, Leung M, Tang IW, Punshon T, Jackson BP, Bidlack FB, Bartell S, Mangano J, Weisskopf MG, Prenatal and Early Postnatal Lead Exposure, Sensitive Periods, and Later Adult Mental Health. Presented at the Annual Meeting of the International Society of Environmental Epidemiology, Atlanta Georgia. 2025
3. **Lin JJ\***, Hickman R, Tang IW, Punshon T, Jackson BP, Bidlack FB, Mangano J, Weisskopf MG, Early life lead exposure associated with higher BMI in late adulthood. Presented at the Annual Meeting of the International Society of Environmental Epidemiology, Santiago, Chile. 2024
4. **Lin JJY\***, Werder EJ, Lawrence KG, Braxton Jackson W, Sandler DP, Dickerson AS, et al. Residential Proximity to Metal-Emitting Industries and Toenail Metal Concentration in the US Gulf States. Presented at the Annual Meeting of the International Society of Environmental Epidemiology, North American Chapter Meeting, Corvallis, Oregon, USA. 2023
5. **Lin JJY\***, Kuiper JR, Dickerson AS, Buckley JP, Volk HE, Rohlman DS, et al. Associations of a toenail metal mixture with attention and memory in the Gulf long-term follow-up (GuLF) study. Presented at the Annual Meeting of the International Society of Environmental Epidemiology, North American Chapter Meeting, Corvallis, Oregon, USA. 2023
6. **Lin JJY\***, Koffman L, Tehrani M, Chen R, Sandler D, Lawrence K, Jackson WB, Ramachandran G, Engel L, Rule A, Reliability of low-mass toenail samples as biomarkers of chronic metal exposure. Presented at the Annual Meeting of the International Society of Exposure Science and Environmental Epidemiology, Lisbon, Portugal. 2022

#### Posters

1. **Lin JJY\***, Koffman L, Tehrani M, Chen R, Sandler D, Lawrence K, Jackson WB, Ramachandran G, Engel L, Rule A, Reliability of low-mass toenail samples as biomarkers of chronic metal exposure. Presented at the 34<sup>th</sup> Annual Meeting of the International Society of Environmental Epidemiology, Athens, Greece. 2022
2. **Lin JJY\***, Tehrani M, Chen R, Heaney C, & Rule A, Characterizing Spatiotemporal Variability in Airborne Heavy Metal Concentration: Changes After 18 Years in Baltimore, MD. Presented at the Annual Meeting of The Health Effects Institute, Washington, DC. 2022
3. **Lin JJY\***, Rule A, Karion A, Koehler K, & Levy-Zamora M., Evaluation of a Low-cost Methane Sensor for Environmental Monitoring in an Urban Setting. Presented at the Annual Meeting of the International Society of Exposure Science Conference. Virtual Meeting. 2021
4. **Lin JJY\***, Tehrani M, Chen R, Heaney C, & Rule A., (2020). Characterizing Spatiotemporal Variability in Airborne Heavy Metal Concentration: Changes After 18 Years in Baltimore,

MD. Presented at the Annual Meeting of International Society of Exposure Science Conference. Virtual Meeting. 2020

## PROFESSIONAL SERVICE

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### **Peer Reviewer – Scientific Journals**

Environmental Health Perspectives, Science of the Total Environment, American Journal of Epidemiology, Journal of Exposure Science and Environmental Epidemiology, Exposure and Health, Environmental Advances, Nicotine and Tobacco Research, Environmental Research, Environmental Science and Pollution Research

### **Conference Abstract Reviewer**

2023, 2024 – International Society for Environmental Epidemiology, conference abstracts for annual meeting

**Organizer and Co-Chair, Symposium:** *Biomonitoring for Environmental Metal Exposures*. International Society for Exposure Science (ISES) Annual Meeting, Chicago, USA. 2024.

## PROFESSIONAL SOCIETIES AND MEMBERSHIPS

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2020 – present, International Society for Exposure Science

2021 – present, International Society for Environmental Epidemiology